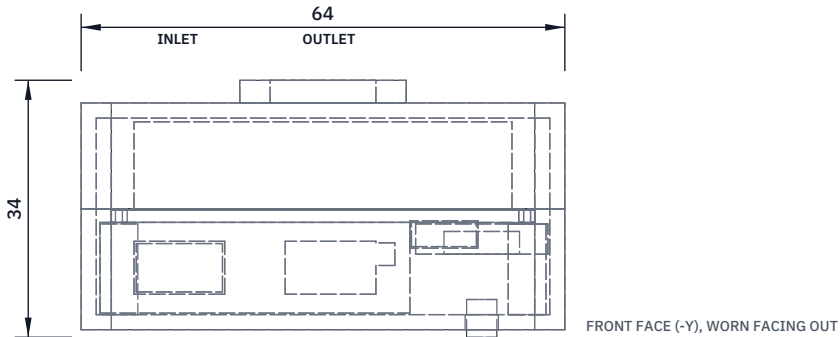


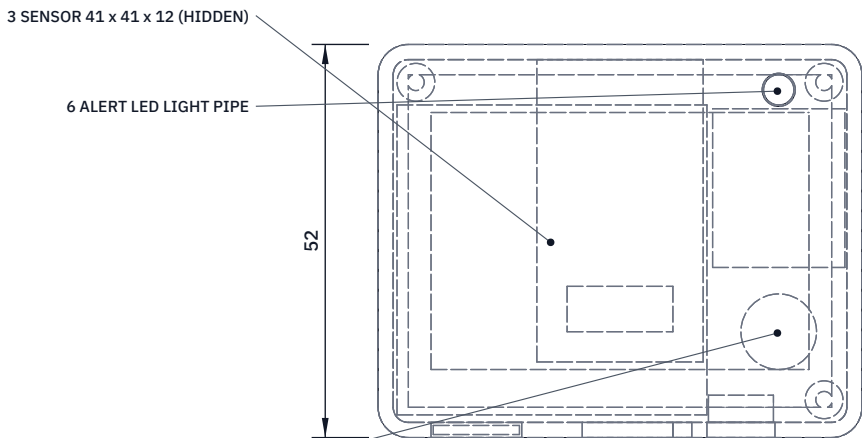
CONCEPT, NOT FOR FABRICATION

DustBadge · github.com/BoujeeEnjinia1701/dustbadge

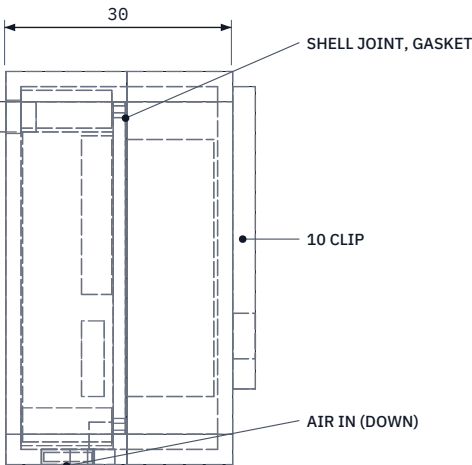
REV	DESCRIPTION	DATE	BY
P1	Preliminary GA for TRL 3 (from cad/src/model.py)	2026-09-25	AC
P2	2,000 mAh cell, 11.5 thick; sun use rule (DBG-DDR-002)	2026-09-25	AC
P3	Layout and labels tidied	2026-09-25	AC



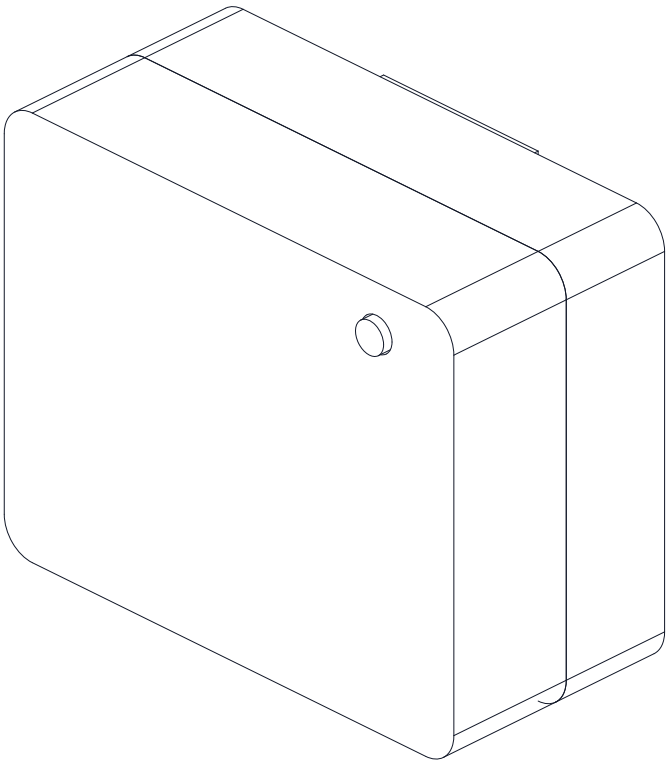
TOP VIEW
Scale 1:1; looking down (along -Z)



FRONT VIEW
Scale 1:1; looking at the front (along +Y)



RIGHT VIEW
Scale 1:1; looking at the right side (along -X)



ISOMETRIC VIEW
Not to scale; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS AND INTERFACES (MM)

Shells 64 x 52 x 30; 34 deep with clip; wall 2; joint 14 from rear
Sensor SPS30 class 41 x 41 x 12, ports down onto two slots
Slots 12 x 7: inlet (screened) at X -19, outlet at X 1
USB-C opening at X 16; humidity vent at X 8; all in the bottom face
Cell 50 x 34 x 11.5, 2,000 mAh, protected, with thermistor
Three M2 bosses; carrier board 56 x 44 rests on them
Mass 119.8 g (DBG-CAL-001); inlet 242 from nose and mouth when worn
Not intrinsically safe; not for gassy mines or explosive atmospheres
Wear shaded when ambient is above 40 °C in full sun
Third-angle; front view from -Y; origin at the badge center, rear face Y = 0

DUSTBADGE

github.com/BoujeeEnjinia1701/dustbadge

General arrangement

DRAWING NO. DBG-DWG-001			REV P3	SHEET 1 of 1	SIZE ANSI B
SCALE 1:1	UNITS mm	PROJECTION 	DATE 2026-09-25		DRAWN Amish Chadha
MATERIAL / NOTES PETG shells; bought-in parts per bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com					Generated from cad/src; do not edit by hand